BEFORE THE
NEW YORK STATE
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service.	)	Case Number: 19-G-0309
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a National Grid for Gas Service.	)	Case Number: 19-G-0310

REBUTTAL TESTIMONY OF BOB WYMAN

September 18, 2019

Introduction

Q. What is your name?

A. I am commonly known as Bob Wyman. My formal name is Robert Mark Wyman.

Q. Are you the same Bob Wyman who submitted direct testimony in this case?

A. Yes.

Q. What is your purpose in submitting this testimony?

A. I wish to comment on several aspects of the direct testimony submitted by other parties in this case. My rebuttal testimony is intended to ensure a more complete record of the relevant issues and to advocate for policies that will be more fair, more in the public interest, and more likely than those proposed by the Company to conform to existing State and New York City laws and policies.

The issues that I wish to discuss include:

- Staff opposition to the proposed "Expanded Geothermal Demonstration Pilot Program"
- New York City's testimony concerning potential stranded asset risk.

Staff Opposition To Expanded Geothermal Demonstration Program

Q. When and why was the"Geothermal Demonstration Pilot Program" begun?

A. Accepting "Staff's recommendation that the Companies [KEDNY, KEDLI] should be required to explore the use of geothermal technologies to improve sustainability and overall system efficiency by displacing peak gas consumption rather than by address-

ing those needs through the addition of pipeline capacity,"[1] the Commission's December 16, 2018 order directed that KEDLI should "implement a pilot program to demonstrate geothermal heating and cooling as an alternative to either new or existing firm or interruptible gas service."

As ordered, KEDLI initiated a Geothermal Demonstration Project "to test shared-loop ground-source heat pump ("GSHP") systems as a cost-effective heating and cooling alternative to natural gas infrastructure in the Glenwood Village community of Riverhead, New York."[2].

Q. Has the Geothermal Demonstration Project been successful?

A. The Companies testify that the project was, in fact, a success and has proven that "geothermal technology has the potential to play an important role in the suite of NPAs called upon to sustainably meet customer heating requirements."[3] Based on an analysis of program results, the company estimates that "each geothermal customer could realize an average annual energy cost savings of between $1,000 and $1,500 and nearly 6.75 metric tons of CO_2 emission reductions from the displacement of approximately 800 gallons of heating oil."[4] The Companies have not estimated the potential savings when natural gas is the alternative fuel.

Q. Please summarize the Companies' proposal for an Expanded Geothermal Demonstration Project.

A. The Companies propose to expand the existing program to include both KEDNY and KEDLI. They propose to own and operate the installed equipment and to charge participating customers a fixed monthly fee of approximately $440 per year to recover

[1] CASE 16-G-0058, *ORDER ADOPTING TERMS OF JOINT PROPOSAL AND ESTABLISHING GAS RATE PLANS*, Dec 16, 2016. pp. 121-122

[2] The Future of Heat Panel, pp. 50-57

[3] The Future of Heat Panel, pp. 52.

[4] The Future Of Heat Panel, pp. 53.

the costs of engineering, design and installation.

Q. Why does Staff oppose expansion of the Geothermal Demonstration Project?

A. In it's testimony,[5] Staff provides three reasons for opposing expansion:

1. "The existing Geothermal Demonstration Project is just entering the data analysis phase and has not yet been proven to be cost-effective."
2. "KEDNY and KEDLI have not demonstrated that utility ownership of a geothermal system would not adversely affect third-party market participants that may be able to provide additional affordable alternatives."
3. "[T]he Companies have not performed a solicitation, such as for a non-pipeline alternative, seeking third-party developers to demonstrate that there is a need for utility ownership."

Q. Is Staff's first objection, concerning un-proven cost-effectiveness compelling?

A. No. The existing Geothermal Demonstration Project, because of its small size and limited operational experience, cannot be expected to have already produced compelling cost-effectiveness data.

It should be recognized that the existing demonstration has only explored the utility of shared geothermal loops in one of a very large number of housing stock types.[6] Even if this one project had been shown to be either cost-effective or not cost-effective, the results from such a limited demonstration would not actually be useful in assessing the wisdom of expanding such efforts to other types of housing stock.

In order to usefully determine if, where, and when shared loop geothermal systems are cost-effective, it will be necessary to pilot such systems in a wide-variety of con-

[5] DPS Staff, *Staff Efficiency & Sustainability Panel*, pp. 52

[6] i.e. The first pilot was installed for a cluster of manufactured homes. While it is important to study this type of housing, such housing comprises only a small percentage of the housing which exists in the Companies' service territory.

ditions and over periods of time long enough to ensure that a good sampling of operational results during a variety of weather patterns can be had.

Ideally, the Company would, as part of their proposed expansion, commit to ensuring that additional demonstrations would be sited in such a way as to explore cost-effectiveness in a variety of situations not yet explored. Some candidates might include:

- Multi-family housing
- Row houses in dense urban environments
- Mixed-use systems which combine commercial heating/cooling loads with residential heating/cooling loads.
- Systems that rely on river, lake, ocean, sewage or waste water as a heat sink/source rather than ground water
- Buildings currently using gas or clusters of buildings which include one or more currently using gas.

While Staff's concern for cost-effectiveness is undoubtedly well-intentioned, restricting expansion of this program at this time prevents the gathering of precisely the cost-effectiveness information that Staff rightly and properly seeks.

Q. Is Staff's second objection, concerning potential interference with the competitive market compelling?

A. Yes and No. Certainly, we should all be concerned whenever a regulated utility seeks to expand its scope into a market which might otherwise be served by others. However, in this case, it is clear that there are currently virtually no "third-party market partipants" whose efforts might be "adversely affected" by an expansion of the existing Geothermal Demonstration Program. Thus, it is too early for this concern to

command our actions in this case.

In fact, the experience and learning that would come from an expanded program might provide the knowledge needed in order to convince third-parties to get into this market. The more successes the Companies accumulate and the more evidence they provide that shared geothermal systems cost-effectively address market needs and that such systems can be profitably owned and operated, the more likely we are to need to address the concern expressed by Staff. On the other hand, a clear demonstration by the Companies that investment in such systems is not prudent, the more likely we are to find that third-parties don't waste their efforts entering a market that holds no promise. Alternatively, the Companies' inability to make such systems work might identify specific issues that others would then address using innovative methods.

Q. How should we address Staff's concern for interference in competitive markets?

A. While there are virtually no third-party participants in the market for shared geothermal systems, there is a small but growing number of companies that currently provide geothermal systems for individual buildings or for collections of adjacent buildings (e.g. Campus-wide systems) having a single owner. The entrance of the Companies into this market for single-owner buildings could have dramatically negative impacts. Thus, It would be very appropriate, in fact essential, to prohibit the Companies from competing with, or interfering with, these existing market participants. The Companies should be restricted to providing shared systems which serve buildings owned by multiple owners.

Even though there are few third-parties in the shared geothermal market today, it is possible that the Company might use means available to it as a regulated utility

to offer systems at rates unachievable by other potential market participants. This would tend to discourage potential third-parties from considering market entrance. (For instance, if the Companies were allowed to recover the costs of such projects through the ratebase shared by all of its other customers, it might be able to amortize costs over periods of time longer than is possible for others while also allocating costs to program non-participants.) Thus, the Companies should be required to ensure that the costs associated with installing systems in the demonstration program will be recovered exclusively from participants in the demonstration program.

Additionally, it would be reasonable to clearly state that approval of an expansion of the demonstration program in this rate case will not imply that future rate cases might authorize continued growth of the program into something other than a demonstration. During each future rate case, careful consideration should be given to determining if continued market participation by the Companies is likely to provide an unwanted interference with the development of a competitive market for shared geothermal systems. In fact, the Companies should be warned that if, at some future time, it is found to be undesirable for them to continue providing the services installed during demonstrations, they may be required to sell-off to third-party market participants any previously installed systems.

Q. Is Staff's third objection, concerning the lack of a solicitation compelling?

A. Somewhat. While the benefits of open solicitations need not be enumerated, it should be recognized that one currently un-stated goal of these demonstrations should be to assist the Companies, the Legislature, and the Public Service Commission in determining whether or not is practical and desirable for utilities to engage more fully in the shared geothermal market. This is particularly important given the expected future decline in natural gas use and the potential reduction in opportuni-

ties for the Companies to either invest in natural gas infrastructure or to continue the employment of those currently on their staffs.

Some argue, as I do, that the Companies should be permitted to shift their focus from the expansion of their natural gas infrastructure to a focus on district heating systems, preferably shared geothermal systems. In brief, while they may stay in the business of providing and operating "Pipes in the Ground," they should learn to **Pump Heat, Not Gas** and become "Thermal Utilities." This transition is necessary, in part, since if the Companies cannot find an alternative to natural gas infrastructure expansion, then they can only be expected to do all in their power to thwart efforts to reduce the consumption of natural gas and thus compromise achievement of the goals established by the Climate Leadership and Community Protection Act as well as NYC Local Law 97-2019. At the same time that we seek a managed process of decapitalization of the Companies' gas assets, we should also ensure that they have a means to profitably deploy their investors' equity through a recapitalization based on district heating assets. Balancing decapitalization with recapitalization is necessary to ensure that all stakeholders interests are best served.

Given that shared geothermal systems are not simply "things we want installed," but also potentially core elements of the Companies' future business model, we should ensure that the Companies are not limited in their ability to retrain existing employees[7] or contractors in order to install the demonstration systems. Any experience they gain in doing so will inform future decisions not only of the Company but of the Commission, Staff and parties to future rate cases.

[7] Ideally, the various unions whose members are employed by the Companies, or their contractors, would become active participants in planning, developing and implementing such retraining programs. Similar unions in Europe have accumulated extensive experience in providing properly trained labor for such projects. One hopes that some knowledge transfer could be facilitated.

Q. What should happen once the demonstrations deliver useful data?

A. Given that district heating systems and shared geothermal systems have proven to be successful and popular in Europe, in areas not that different the urban and near-urban sections of the Companies' territory, it is expected that the demonstration pilots will provide convincing evidence that such systems can provide a viable alternative to gas and oil based systems here. At that point the Commission will need to answer some important questions, including:

- Should utilities be allowed to install shared geothermal systems in non-demonstration programs?
- Should third-party providers of shared geothermal systems be regulated?

I suggest that the answer to both of these question is "Yes." I also recognize that these are not questions that can be resolved in this rate case.

Q. Why should we allow utilities to invest in shared geothermal and district heating systems?

A. As stated elsewhere, I believe that we must provide a pathway to future profitability for the existing gas utilities who have served our interests in the past. As we require that utilities reduce their investment in gas expansion, we should provide them with an alternative investment that will allow them to provide their shareholders with confidence that their investments will continue to afford reasonable returns in the future. We should also show a pathway to a future business model that will avoid the high cost of capital and low credit ratings that are typical of businesses in decline and which would create significant burdens for any remaining gas customers.

Q. Why should the shared geothermal and district heating markets be regulated?

A. Historically, it has been found that the management and development of distribution systems serving multiple customers, such as those for water, gas, electricity, tele-

phone calls, cable TV, and steam, present complex issues that require resolution by a neutral third party dedicated to the public interest. Thus, these "public services" are now regulated by the Public Service Commission. In time, I'm confident that we'll find that district heating systems, including shared geothermal systems, will present much the same kind of issue that has justified regulation of other shared distribution systems in the past. Given this, I suggest that it would be appropriate to begin now the discussion of how such shared assets should be regulated and when such regulation should begin.

While I suggest that PSC regulation of shared systems will probably be found to be necessary, I do not propose that such regulation should be considered appropriate for systems installed for the benefit of individual owners.

New York City's testimony concerning stranded asset risk

Q. To what testimony do you refer?

A. The "New York City Policy Panel" testimony, starting on page 25 and continuing until page 31.

Q. Do you agree with the City's proposals concerning stranded asset risk?

A. Yes. But, I think the City understates that significance of the stranded asset risk. They say:

> "Given the goals of deep decarbonization and the clean energy transition, the composition of utility infrastructure will change over time. In particular, the need for the Companies' extensive gas infrastructure *may* diminish." (emphasis added.)

While I recognize that the City may have sought to be "gentle" in its illumination of

the issue, I believe the reality is that, given current law and policy, it is a certainty that the need for the Companies' extensive gas infrastructure *will* diminish.

We do stakeholders no service by sugar coating the implications of current law or shying away from stating clearly the consequences of our actions to ensure increased energy efficiency and a reduction of greenhouse gas emissions.

Q. What is it about their proposal that you support?

A. I strongly support the City's call for studies of stranded asset risks and for their proposing the preparation of:

> "An assessment of whether it is technically feasible to not replace the leak-prone pipe in certain neighborhoods without affecting other customers, such as at the end of pipeline segments or at the edges of the Company's gas service territory;"

While the City has provided a number of useful suggestions for similar projects to be studied, I suggest that the list should be expanded to include a study of the Non-Essential Gas Asset Retirement program recently described by Central Hudson.[8]

That program's goals are summarized as follows:

> 1. Non-Essential Gas Asset Retirement
>
> Central Hudson's TMA [Transporation Mode Alternative] projects are designed to strategically retire high-cost natural gas infrastructure by seeking high efficiency electric alternatives to replace natural gas end uses in customer's homes. For each identified area, performance will be measured based on the retirement of all prescribed gas services and mains. The conversion approach will vary by customer based on existing baseline condi-

[8] Cases 17-E-0459, 17-G-0460, *Central Hudson Gas & Electric Corporation's Non-Tariff Implementation Plan & Compliance Filing for Non-Pipe Alternatives: Three Transportation Mode Alternatives,* June 21, 2019

tion, but will primarily focus on high efficiency cold climate air-source heat pumps and electric heat pump water heaters. Other appliance types will be considered based on the existing natural gas end uses for each home.

CHGE may, in pursuing this program, provide important learning that could inform the Companies in pursuing similar efforts. Of course, it would also make sense for the Companies to consider geothermal systems, shared or not, as alternatives to the air-source systems which are the focus of CHGE's program.

Q. Does that conclude your rebuttal testimony in this case?

A. Yes.